

Wind tunnel testing and frequency domain buffeting analysis of a 5000 m suspension bridge

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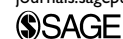
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Abstract

Buffeting's importance is highlighted in the case of super-long bridges. It remains unclear how buffeting will perform and whether the traditional methods still apply when bridge's span is approaching its upper limitation. To figure out this, a full aeroelastic model testing has been carried out with the span arrangement 2000 + 5000 + 2000 m. A comparison with the existing bridges reveals the similarities and differences. It shows vertical and torsional buffetings are important for a 5000 m bridge while lateral response is almost static. Fundamental modes contribute most kinetic energy. With the increasing of wind speed, the aerodynamic coupling effect becomes prominent while the structural coupling effect keeps limited. Afterwards, four analytical methods, namely SRSS and CQC, with and without self-induced forces, are used for the buffeting displacements. The calculation indicates the multi-mode (CQC) analysis is still applicable on a 5000 m bridge but the mode-by-mode (SRSS) method is risky since the neglect of aerodynamic coupling effect. Parameter analysis shows aerodynamic admittance function and force coherence are decisive while the cross terms are of secondary importance. Since all data is experimentally measured, the reliability of analytical results can be ensured.

Keywords

5000 m suspension bridge, aerodynamic admittance function, aerodynamic coupling, bridge buffeting, full aeroelastic model testing, joint acceptance function, quasi-steady theory, self-induced force

Introduction

With the prevalence of cross-strait projects, bridge's span length witnessed a rapid increase since late last century. The enlargement of span length reduces overall stiffness, making the bridge more vulnerable to fluctuating wind loading. Although buffeting does not have an instability character like flutter, excessive buffeting responses will give rise to fatigue and issues with user's comfort. Research shows the upper limitation of suspension bridges is 5000 to 5900 m (Ge and Xiang, 2006, 2008). Two problems relating to buffeting need to be answered for the further development as, (1) what kind of role turbulence and buffeting will play in that super-long case, and (2) whether a traditional method can still be used to analyze the buffeting in that case.

The analysis of bridge's buffeting can be tracked back to 1960s when Davenport (1961, 1962) firstly introduced stochastic vibration theories into wind engineering. However, only aerodynamic damping is accounted in the early work. Scanlan (1977, 1978) therefore developed Davenport's theory in the view of self-induced force. The total aerodynamic forces are the superposition of buffeting forces and self-induced

forces. Although in early works, Scanlan emphasized the contribution of self-induced force but ignored the influence of upstream turbulence scales, aerodynamic admittance functions were later adopted (Scanlan, 1993).

Various empirical approaches were further developed for buffeting analysis, among which most are the mathematical applications of Davenport's or Scanlan's methods. For instance, Jain et al. (1996) accounted the aerodynamic coupling effect and carried out multi-mode analysis in the domain of reduced time $s = Ut/B$. Minh et al. (1999) proposed a complex-mode-based

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method, while Xu et al. (2000) conducted the buffeting calculation of the 1377 m-spanned Tsing Ma Bridge by finite element method and pseudo-excitation method.

In the background of ultra-long bridge study, a full aeroelastic model was built (Ge et al., 2018) with a total length of $2000 + 5000 + 2000$ m. Its buffeting responses are presented in this paper followed by the frequency-domain calculations using experimental data. The paper is arranged as follows. The structural details of a 5000 m main span bridge and its full aeroelastic model testing are given in Part 2. The comparison versus existing bridges classifies their similarities and differences. In the next part, frequency domain methods are reviewed. A detailed discussion reveals the aerodynamic parameters via sectional model tests. Based on wind loading and responses' combination, four classical methods are used for buffeting analysis, which are, SRSS and CQC, with and without self-induced forces. Sensitivity analysis helps investigate the influence of aerodynamic parameters. Conclusions are drawn in the final. All the data used for computation are experimentally measured, which ensures the reliability of analysis.

Full aero-elastic buffeting testing

Structural layout

The mechanical advantage of suspension bridges makes them have a larger spanning capacity. Ge and Xiang (2006) proposed the maximum span length can reach 5000 to 5900 m if steel main cables are used. When the span length exceeds 5000 m, aerodynamic problems turn to be dominant since both wind loading and structure are of low frequencies. However, the longest bridge studied so far is the Messina Strait Bridge (main span: 3300 m; Diana, 2003). Since it remains unexplored what a role wind loading will play once the span length approaches this limitation, a 5000 m bridge has been designed and its full aeroelastic model constructed and tested. The span arrangement is $2000 + 5000 + 2000$ m (Figure 1(a)). Wind tunnel testing has been carried out in the TJ-3 Wind Tunnel, which is 15 m wide, 2 m high, and 14 m long. The full bridge aeroelastic model has a geometric scale ratio of 1:620 so that the model length is 14.7 m (Figure 1(b)).

For a better flutter performance, a central-slotted girder was recommended in the preliminary study (Shao et al., 2011). The girder's total width is 80 m (Figure 1(a)). Each side box is connected to two main cables. The cross section is simplified as a 40 m central slot and two 20 m side-boxes. Discussions on flutter performance can be found in Xia (2020) while this paper focuses on the buffeting performance.

Figure 1(c) to (e) shows the main components of the full aero-elastic model. Girder's flexural and torsional stiffnesses, with a scale ratio of $1:620^5$, are provided by aluminum cross beams and steel main beams. They have rectangular sections of 2.3×0.5 mm, and 0.4×2.1 mm (Figure 1(d)). These beams are surrounded by the green foam of which the density is 0.03 g/cm^3 . The 0.17 mm thick ABS plate ensures the surface is smooth (Figure 1(c)). Resulting from the large span length and the small geometric scale ratio of 1:620, the scale ratio of mass is $1:620^2$ and that of mass moment of inertia is $1:620^4$. After the subtraction of these mass, some additional mass of plumbum wires are added. The span-wise consecutive main beam and the 1mm gap of foam ensures there is no additional stiffness. As given in Figure 1(e), high-strength steel wires of 0.1 mm diameter are used as suspenders and main cables. The full-scale main cable's area is 1.872 m^2 with an axial stiffness scale ratio of $1:620^3$. The deficient mass and drag force of cable systems are simulated by a set of brass tubes of which the density is 8900 kg/m^3 . This small and elaborate model has been carefully aligned before the experiment.

Free vibration properties and validation

Natural frequencies and modal shapes, as the comprehensive results of mass and stiffness distributions, are the basics of dynamic analysis. In Figure 1(b), 11 span-wise measurement points were set with vertical, torsional, and lateral displacements recorded. Laser transducers simultaneously measured the free vibration displacements after short-time impulses. A global curve fitting algorithm (Richardson and Formenti, 1985) leads to the free vibration properties in Figure 2. The data in the brackets are the frequencies calculated by finite element model (FEM). A good matchup of the tested frequencies with those design frequencies demonstrates the reliability of the model. After the verification of experimental data, the equivalent masses (or mass moments of inertia) are calculated using Ansys 15.0.

In Figure 2, "*I-S-L*" denotes the first symmetric lateral mode and so forth. The subscript "*model*" means model scale while "*bridge*" is for full-scale bridge. $p(x)$ and $h(x)$ are the lateral and vertical displacements and $\alpha(x)$ is the torsional displacement. To make the translational displacements, $p(x)$ and $h(x)$, comparable with the rotational ones, $\alpha(x)$, the torsional displacements are multiplied by deck width B in Figure 2. The corresponding scale ratio of frequency is 24.9:1 and of wind speed is 1:24.9. The equivalent mass (per meter) has a scale ratio of $1:620^2$, while the scale ratio of equivalent mass moment of inertia (per meter) is $1:620^4$. Models'

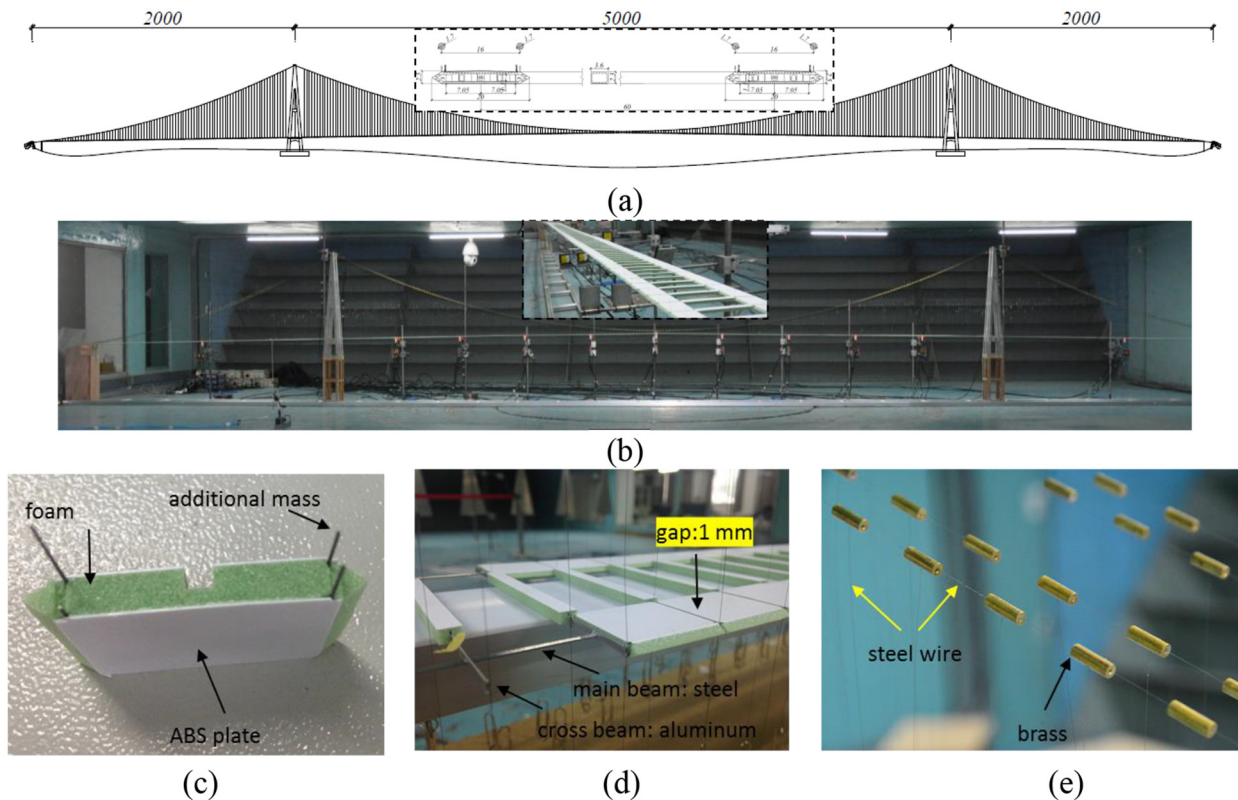


Figure 1. Structural layout of: (a) 5000 m bridge, (b) full aeroelastic model (unit: m), and (c–e) details of model.

damping ratios are in the range of 0.5% to 1.5%. During the wind tunnel testing, the second symmetric vertical mode (2-S-V) hasn't been excited. The property of 2-S-V from FEM would be directly employed in the later calculation.

A 5000 m bridge's free vibrations along three directions, that is, vertical, lateral, and torsional, are well separated. Especially for the 1st modes, most modal shapes are dominated by the displacement along one direction. The fundamental symmetric modal shape of a normal size bridge is of a half-wave, such as the 1st lateral mode in Figure 2(a). However, the 1st symmetric vertical and torsional modes are of three alternating half-waves. This case cannot be observed on a normal bridge but is a common phenomenon in super-long cases. In this super-long case, the pylon must be strong which in some sense restrains the girder in the near-pylon region, making its displacement out of phase with the midspan range. A specialized terminology, "first-third symmetric mode," was given by Diana (2003).

Experiment results

Considering buffeting is a long-term dynamic problem and the general angle of attack is 0° in a strait terrain,

the testing was carried out at that angle. In most bridges' testing (e.g. Ge et al., 2010, 2017), turbulence intensities at deck height are 14% according to the design code. A series of mobile spires were fabricated with heights and projected areas adjustable (Figure 3(a)). A detailed discussion of wind field in the wind tunnel can be found in Ge et al. (2018). The combination of these spires with roughness makes wind spectra match the von-Karman spectrum (Figure 3(b)) and the wind profiles meet the exponential law. Model's buffeting responses were synchronously measured with the upstream flows. In buffeting testing, three turbulence configurations offered intensities increasing from 10% to 14%.

Overall buffeting response. Figure 4 gives the measured buffeting responses when turbulence intensities were 10%, 11%, and 14%. Since there is no lateral constraint in the main span, the girder is relatively flexible along that direction so that the static deformation playing a more important role than dynamics. When the upstream velocity is 51.1 m/s and intensity is 14%, model's lateral deformation is 1/500 of main span length while the RMS value is only 9.9% of that. As a comparison, the vertical RMS response is two times of the static deformation and the torsional ratio is more

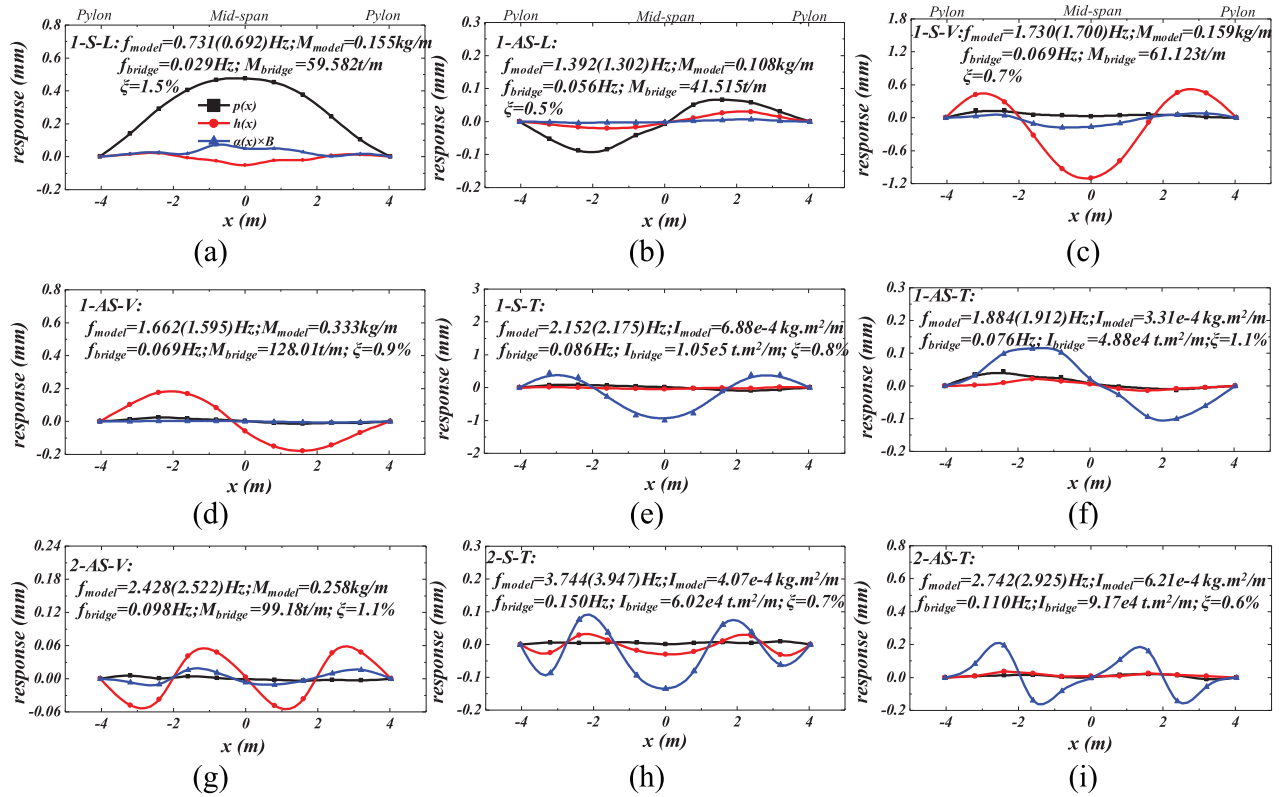
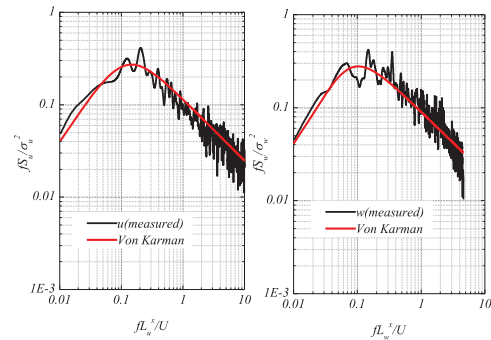


Figure 2. Free vibration properties of a 5000 m bridge model (V: vertical; L: lateral; T: torsional). (a-b): 1st symmetric and anti-symmetric lateral mode; (c-d): 1st symmetric and anti-symmetric vertical mode; (e-f): 1st symmetric and anti-symmetric torsional mode; (g): 2nd anti-symmetric vertical mode; (h-i): 1st symmetric and anti-symmetric torsional mode.



(a)



(b)

Figure 3. (a) adjustable spires and (b) typical wind spectra ($l = 14\%$, $U_{\text{model}} = 1.1 \text{ m/s}$).

prominent. A later discussion of response spectra shows the lateral response is irrelevant with the other two directions. Therefore, the lateral buffeting can be treated as a static phenomenon.

It also shows the mean responses are not sensitive to the turbulence intensities, which is reasonable since mean values are caused by mean wind loading. The symmetry of deck makes the mean moment and lift force coefficients zero so that the mean vertical and

torsional deformations are small. When the bridge wind speed is 51.1 m/s , the mean lateral response can be controlled to the $1/500$ of span length. From another perspective, a wide-slotted deck is necessary since such a deck provides enough lateral stiffness, leading to less lateral deformation. The present ratio of deck width to span length is $80/5000 = 1/62.5$. By contrast, the ratio of Akashi Kaikyo Bridge is $1/56.1$ and of Great Belt Bridge is $1/64.1$.

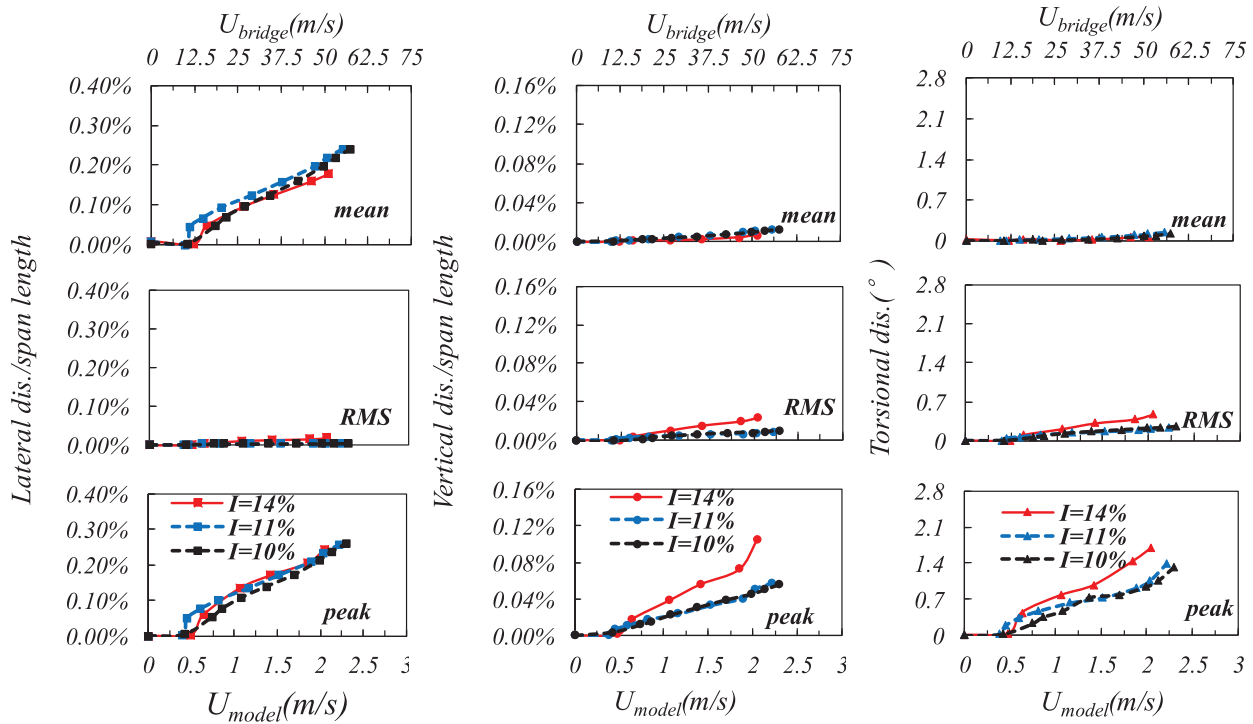


Figure 4. Overall buffeting responses at three turbulence intensities (from left to right are lateral, vertical and torsional responses).

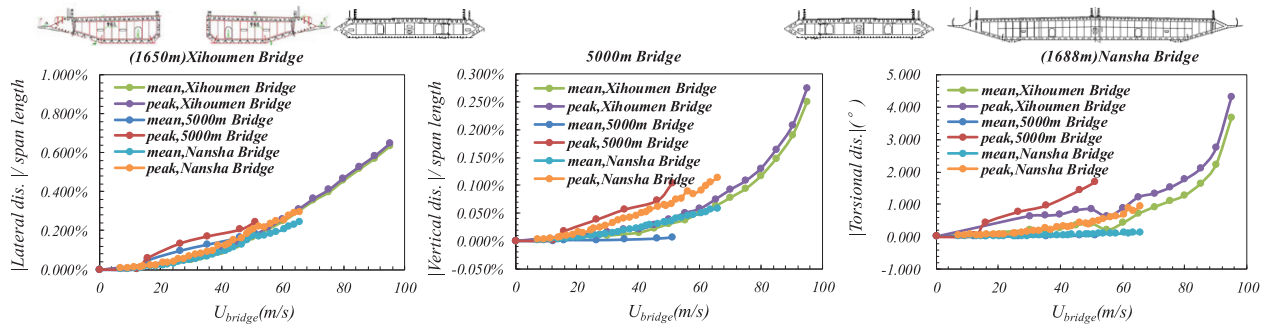


Figure 5. Comparison with existing bridges ($I = 14\%$).

By comparison, the RMS responses are the direct results of turbulence intensities. In the cases of $I = 10\%$ and 11% where intensities are close, the integral sales have a 25% difference because of the spire locations in the wind tunnel (details in Ge et al., 2018). However, the dynamic responses keep close to each other. It implies the more decisive upstream condition is turbulence intensity. To have a practical view, the vertical RMS response of a 5000 m bridge is 1/10000 to 1/5000 of span length when wind speed is 50 m/s and the corresponding torsional RMS approaches 0.45° .

Attentions are paid to the peak responses for bridge design. The peak vertical and lateral responses show that when the wind speed is 51.1 m/s, their ratios to span length are around 1/500 and 1/1000 which is accepted by most wind codes. Transversal responses will not be a big problem for user's comfort. However, the torsional response, in the order of 1.0° , requires more farther considerations.

Comparison with existing bridges. For a practical view, the tested data is compared with the existing ones. Two typical bridges are listed in Figure 5, namely the

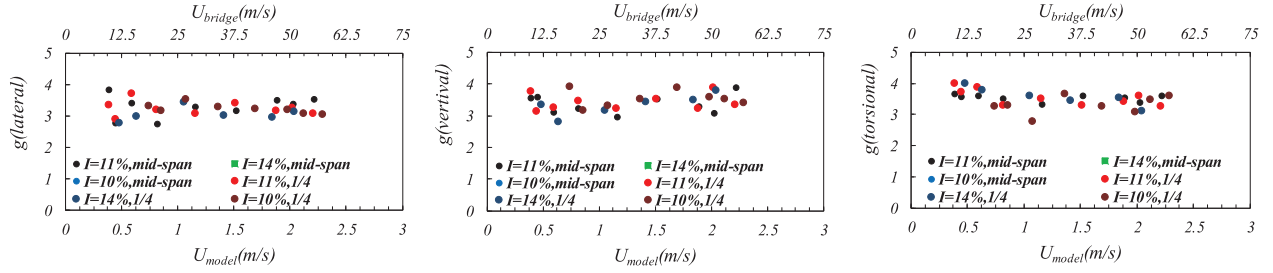


Figure 6. Peak buffeting factors (from left to right are lateral, vertical and torsional factors).

1650 m-spanned Xihoumen Bridge and the 1688 m-spanned Nanshan Bridge, which are the fourth and third longest suspension bridges in service. Their peak and mean values were also tested in Tongji BLWT (Ge et al., 2010, 2017) with the turbulence intensity 14%. Also listed in the figure are their cross sections. The former deck has a width of 36.0 m including a 6.0 m slot, while the later bridge width is 49.7 m.

As Figure 5 shows, three bridges' lateral displacements are comparable with each other, and their mean lateral responses are close to the peak values. When the upstream wind speed is 50 m/s, bridges' peak lateral responses are the 1/500 of span length. Meanwhile, the 5000 m bridge's vertical peak responses are close to Nansha Bridge but larger than Xihoumen Bridge. Since Xihoumen Bridge and Nansha Bridge have a similar span length as well as comparable dynamic properties, the deviations indicate the buffeting responses are highly dependent on the deck configuration.

Unlike the consistencies along vertical and lateral directions, the torsional displacements are different. The 5000 m bridge's torsional response is the order of 1.5° while the others are much smaller when wind speed is 50 m/s. Even for Nansha Bridge and Xihoumen Bridge of similar span length but different deck types, the former's torsional displacement is much smaller. It is indeed caused by the configuration of the central slot. The later discussion shows AAF is one dominating factor and torsional AAF is highly enlarged by the setup of a central slot.

Buffeting's peak factor. The application of a peak factor, g , realizes the estimation of maxima. As given in equation (1), it is defined as the number of standard deviations that separates the peak value from the mean value.

$$\hat{r} = \bar{r} + g\tilde{r}, \quad (1)$$

In the equation, the subscript “^” means the peak value, “-” denotes mean value and “~” is for standard deviation. In atmospheric boundary layer, it is dependent on numerous inputs, including the roughness length, distance from an upstream terrain change and

height. Davenport (1964) indicates the peak response of most structures to gusts can be expected in the range of 3.5 to 4.5 standard deviations of the response in excess of the mean. Figure 6 shows the peak factors for a 5000 m bridge are from 3.0 to 4.0, lower than the recommended range. Although some fluctuations can be observed from the figures, these peak factors are insensitive to turbulence intensity, mean wind speed and longitudinal location. Meanwhile, the lateral gust factors are slightly smaller than the others. This is understandable since the mean value plays a more important role than the fluctuations along the lateral direction.

Effect of signature turbulences. The flow field of a twin-box girder is different with a single box since the windward box's wake will impinge on the leeward deck and cause additional fluctuations. This turbulent wake in the slot has a terminology as “signature turbulence.” Deck's total wind loading is therefore the comprehensive result of upstream turbulence and signature turbulence.

Whereas, it is impractical to separate signature turbulence from the upstream turbulence. To estimate the contribution of signature turbulence, the responses under smooth flow are compared with the turbulent inflow cases (Figure 7). In the former case, namely there is only signature turbulence, the mean response is close to the peak values. It indicates upstream turbulence is the main cause of dynamic responses though signature turbulence may exist. Considering the deck configuration, it is understandable since the central slot is quite wide and the small deck depth makes the wake frequency much higher, thus, far away from the structural frequency.

Typical buffeting response. Figure 8 gives the typical span-wise displacements when turbulence intensity is 14%. Unlike a normal suspension bridge of which the maximum buffeting response occurs at the mid-span or the quarter-span, the RMS of a 5000 m bridge is uniformly distributed in the locations away from the pylons. This means the constraint by the pylon is limited in the 25% of span length. Since the response

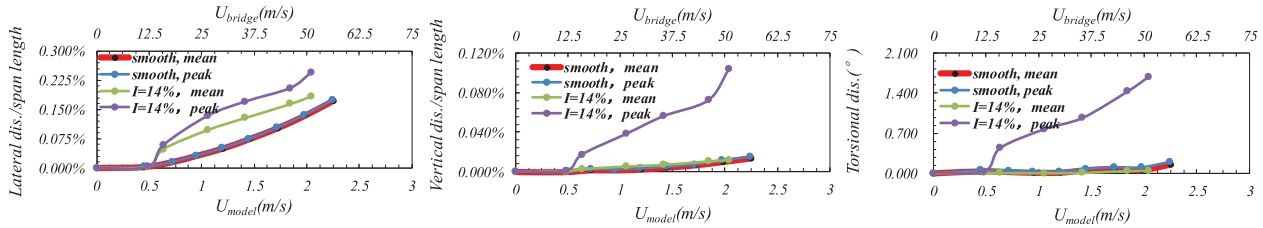


Figure 7. Responses under smooth and turbulent flows at mid-span (from left to right are lateral, vertical, and torsional responses).

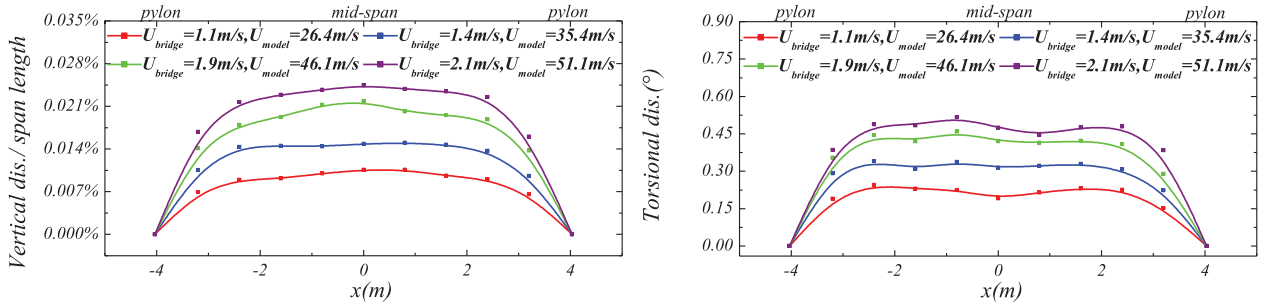


Figure 8. Measured RMS buffeting responses in the main span ($I = 14\%$).

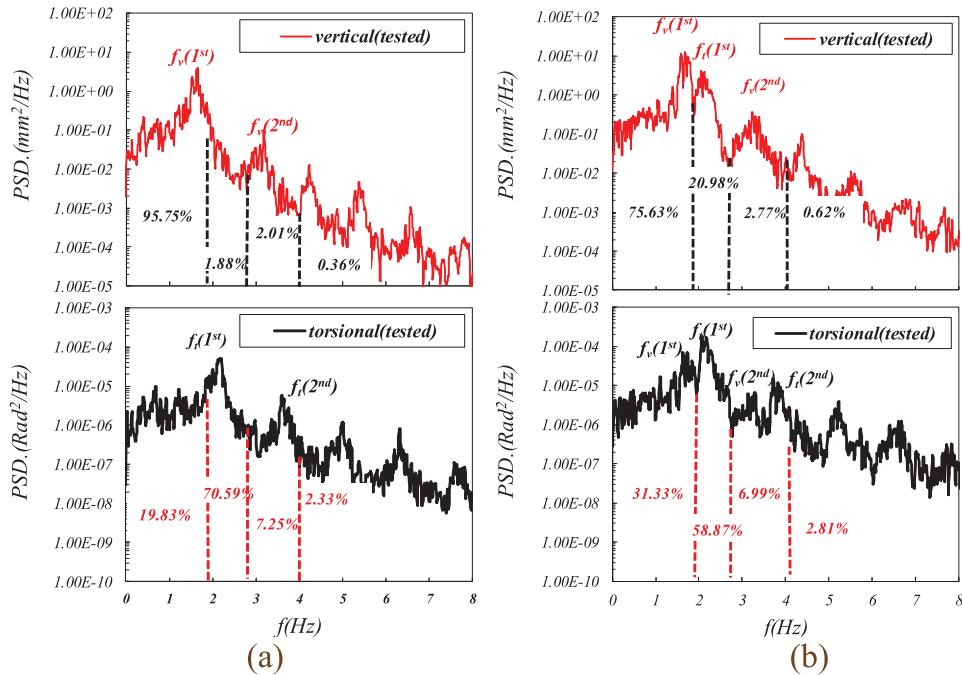


Figure 9. Buffeting response spectra at mid-span ($I = 14\%$): (a) $U_{\text{model}} = 1.1$ m/s, $U_{\text{bridge}} = 26.4$ m/s and (b) $U_{\text{model}} = 2.1$ m/s, $U_{\text{bridge}} = 51.1$ m/s.

magnitudes are similar in most regions, the following discussion will focus on the mid-span.

The displacements at mid-span are given in the frequency domain (Figure 9) for a detailed and clear view. The area underneath means the kinetic energy proportion in that frequency range. The horizontal

axis is divided into four regions with energy percentages labeled. As it shows, most energy concentrates in the neighboring region of the fundamental modes. This phenomenon is more apparent for the vertical displacements. The second mode has some weights along torsional direction. Higher modes' contributions to

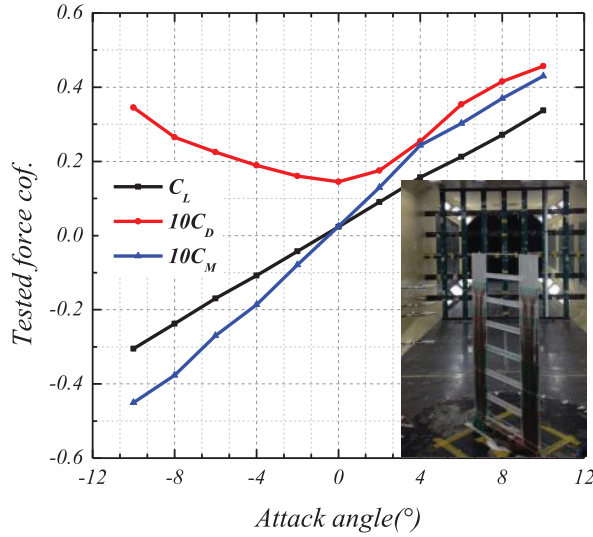


Figure 10. Sectional pressure testing model and mean force coefficients.

both vertical and torsional buffeting responses are negligible ($<3\%$). It also seems higher modes are more important for the torsional displacements.

The energy distribution is more complex at a higher wind speed. In Figure 9(a), only two peaks of 1st and 2nd symmetrical modes are prominent in the region that $f < 4.0$ Hz. When the speed is high (Figure 9(b)), vertical modes' frequencies arise in the torsional spectrum and vice versa. This cross-frequencies account for a large proportion. To be specific, the vertical energy contribution from $f = 1.8$ Hz to 2.8 Hz is $<2\%$ at $U_{\text{model}} = 1.1$ m/s while it rapidly increases to be 20.98% when experimental wind speed is 2.1 m/s. This is the so-called aerodynamic coupling effect caused by aeroelastic effect. It implies the consideration of first two modes (both symmetric and anti-symmetric in other locations) is of enough accuracy. The lateral frequencies are not apparent in the PSD of vertical and torsional responses. Considering the lateral RMS responses in Figure 4 are small, lateral response's contribution can be neglected when computing the vertical and torsional responses.

Frequency domain buffeting analysis

In the above chapter, the buffeting of a 5000 m bridge model as well as the detailed discussion has been presented. The subject of this part is to check the applicability and compare the precision of analytical methods using experimental data.

A review of analytical method

Based on wind loading calculation, frequency-domain buffeting analysis can be divided into two categories,

as Davenport's quasi-steady theory and Scanlan's unsteady modification. Besides, depending on the combination scheme of modal responses, the dynamic analysis is classified into SRSS method and CQC method. Therefore, totally four methods are going to be employed.

Dynamic combination schemes. The principle of frequency-domain analysis is modal superposition. A single mode consists of vertical, lateral and torsional displacements. The total response using normal coordinates and modal shapes reads,

$$\begin{cases} h(x;t) = \sum_{i=1}^n B h_i(x) \xi_i(t) \\ p(x;t) = \sum_{i=1}^n B p_i(x) \xi_i(t), \\ \alpha(x;t) = \sum_{i=1}^n \alpha_i(x) \xi_i(t) \end{cases} \quad (2)$$

where i is the number of free vibration mode ($i = 1-n$), and ξ_i is the normal coordinates.

A multi-degree of freedom (MDOF) system can be decoupled into a set of equivalent single degree of freedom (ESDOF) systems if it is both structurally and aerodynamically uncoupled, which goes,

$$M_i \ddot{\xi}_i(t) + C_i \dot{\xi}_i(t) + K_i \xi_i(t) = Q_i(t). \quad (3)$$

where M_i , C_i , K_i are generalized mass, damping and stiffness. Therefore, the auto- or cross-spectrum of modal response can be calculated as,

$$S_{\xi_i \xi_j}(\omega) = H_i(\omega) H_j(\omega) S_{Q_i Q_j}(\omega), \quad (4)$$

where $H_i(\omega)$ is frequency response function, $S_{Q_i Q_j}$ is the modal loading spectrum and $S_{\xi_i \xi_j}$ is the spectrum of normal coordinate. After the calculation of $S_{\xi_i \xi_j}$, the SRSS and CQC methods are widely used to determine maximum response. As given by equation (5), it goes to SRSS when $i = j$. Otherwise, all cross terms are counted which refers to CQC. In general, SRSS method has enough accuracy. Once the frequencies of interested modes are close to each other, SRSS method gives poor results and the CQC method is expected.

$$\begin{cases} S_{hh}(x; \omega) = B^2 \sum_i \sum_j [h_i(x) h_j(x) S_{\xi_i \xi_j}(\omega)] \\ S_{pp}(x; \omega) = B^2 \sum_i \sum_j [p_i(x) p_j(x) S_{\xi_i \xi_j}(\omega)] \\ S_{\alpha\alpha}(x; \omega) = \sum_i \sum_j [\alpha_i(x) \alpha_j(x) S_{\xi_i \xi_j}(\omega)] \end{cases} \quad (5)$$

Quasi-steady force and buffeting response. The quasi-steady hypothesis implies the aerodynamic forces are

the only functions of instantaneous wind speeds. Deck's fluctuating forces can be expressed by equation (6) under small displacement assumption (Davenport, 1962; Scanlan, 1978).

$$\begin{cases} D_b(t) = \frac{\rho UB}{2} [2C_D(\alpha_0)u(t) + C_D'(\alpha_0)w(t)] \\ L_b(t) = \frac{\rho UB}{2} [2C_L(\alpha_0)u(t) + [C_L'(\alpha_0) + C_D(\alpha_0)]w(t)] \\ M_b(t) = \frac{\rho UB^2}{2} [2C_M(\alpha_0)u(t) + C_M'(\alpha_0)w(t)] \end{cases} \quad (6)$$

In equation (6), D_b , L_b , and M_b are the drag, lift and torque; U is mean wind speed with α_0 the mean attack angle; u and w are the streamwise and transverse wind fluctuations; C_D , C_L , and C_M refer to force coefficients defined by the width of deck B ; the superscript denotes the gradients of coefficients and ρ is the air density.

As turbulence consists of various wave-number components, the coefficients in equation (6) will change under different inflow conditions. An increased wave number refers to a smaller eddy size so that the pressure around deck tends to be less synchronized and a reduction is expected. For an airfoil's reduction, Sears (1941) proposed the aerodynamic admittance function (AAF). Davenport adopted this terminology into bridge analysis. Leaving out the cross spectra, the buffeting force is modified by equation (7), where S indicates force spectra and $|\chi|$ denotes aerodynamic admittance function.

$$\begin{cases} S_{D_b}(\omega) = \left(\frac{\rho UB}{2}\right)^2 [4C_D^2(\alpha_0)S_u(\omega)|\chi_{Du}(\omega)|^2 + C_D'^2(\alpha_0)S_w(\omega)|\chi_{Dw}(\omega)|^2] \\ S_{L_b}(\omega) = \left(\frac{\rho UB}{2}\right)^2 [4C_L^2(\alpha_0)S_u(\omega)|\chi_{Lu}(\omega)|^2 + [C_L'(\alpha_0) + C_D(\alpha_0)]S_w(\omega)|\chi_{Lw}(\omega)|^2] \\ S_{M_b}(\omega) = \left(\frac{\rho UB^2}{2}\right)^2 [4C_M^2(\alpha_0)S_u(\omega)|\chi_{Mu}(\omega)|^2 + C_M'^2(\alpha_0)S_w(\omega)|\chi_{Mw}(\omega)|^2] \end{cases} \quad (7)$$

The introduction of aerodynamic admittance function ensures the force coefficients under smooth flow can be directly used. Afterwards, the integral of modal shape transfers the 2D sectional force to 3D generalized modal buffeting loading. The modal buffeting loading is expressed as,

$$Q_{bi}(t) = \int [D_b(t)Bp_i(x) + L_b(t)Bh_i(x) + M_b(t)\alpha_i(x)]dx. \quad (8)$$

The corresponding spectrum of modal buffeting force goes,

$$\begin{aligned} S_{Q_{bi}Q_{bj}}(\omega) = & B^2 S_{D_b} |J_{p_i p_j}|^2 + B^2 S_{L_b} |J_{h_i h_j}|^2 + S_{M_b} |J_{\alpha_i \alpha_j}|^2 \\ & + B^2 \sqrt{S_{L_b} S_{D_b}} (|J_{h_i p_j}|^2 + |J_{p_i h_j}|^2) + \\ & + B \sqrt{S_{L_b} S_{M_b}} (|J_{h_i \alpha_j}|^2 + |J_{\alpha_i h_j}|^2) \\ & + B \sqrt{S_{D_b} S_{M_b}} (|J_{p_i \alpha_j}|^2 + |J_{\alpha_i p_j}|^2), \end{aligned} \quad (9)$$

where $|J|^2$ is the joint acceptance function (JAF), as the double integral of the modal shapes and span-wise coherence, $R_{F_1 F_2}(x_1, x_2; \omega)$. Since the quadrature spectrum of buffeting force is small under isotropic flows, the cross-coherence can be replaced by root coherence as, $R_{F_1 F_2}(x_1, x_2; \omega) = |S_{F_1 F_2}(x_1, x_2; \omega)| / \sqrt{S_{F_1 F_1}(x_1, x_1; \omega) S_{F_2 F_2}(x_2, x_2; \omega)}$.

After the transformation of the sectional force to normal coordinates using JAF by equation (9) and the decoupling of MDoF system as well as the determination of ESDoF response by equations (3) and (4), the schemes of SRSS or CQC can be used for the dynamic combination for multi-mode responses

Modification of unsteady effect. The movement of a bridge deck changes the boundary conditions of the flow and affects the fluid force, which in turns influences the dynamic equilibrium of the deck. Therefore, the aerodynamic force becomes not only the function of flows but also motion related. For a thin airfoil in incom-

pressible flow where potential flow theory applies, Theodorsen (1949) showed self-induced forces are the functions of deck motions and their first and second derivatives. Although it is infeasible to theoretically develop such expressions, Scanlan (1971) proposed that self-induced forces on a bluff body can be treated as the linear functions of displacements and their first order derivatives. Using a set of flutter derivatives, self-induced forces are modeled as equation (10).

$$\begin{cases} D_{se}(t) = \rho U^2 B \left[KP_1^* \frac{\dot{h}}{U} + KP_2^* \frac{B\dot{\alpha}}{U} + K^2 P_3^* \alpha + K^2 P_4^* \frac{h}{B} + KP_5^* \frac{\dot{p}}{U} + K^2 P_6^* \frac{p}{B} \right] \\ L_{se}(t) = \rho U^2 B \left[KH_1^* \frac{\dot{h}}{U} + KH_2^* \frac{B\dot{\alpha}}{U} + K^2 H_3^* \alpha + K^2 H_4^* \frac{h}{B} + KH_5^* \frac{\dot{p}}{U} + K^2 H_6^* \frac{p}{B} \right], \\ M_{se}(t) = \rho U^2 B^2 \left[KA_1^* \frac{\dot{h}}{U} + KA_2^* \frac{B\dot{\alpha}}{U} + K^2 A_3^* \alpha + K^2 A_4^* \frac{h}{B} + KA_5^* \frac{\dot{p}}{U} + K^2 A_6^* \frac{p}{B} \right] \end{cases} \quad (10)$$

where P^* , H^* , and A^* are flutter derivatives and $K = \omega U/B$ is the reduced frequency. Using the normal coordinates in equation (2), self-induced force (lift for instance) can be modeled by,

factors in these aerodynamic matrices are the functions of c_{ij}^{se} and k_{ij}^{se} . Therefore, system's equivalent damping matrix ($[C]-[C^{se}]$) and stiffness matrix ($[K]-[K^{se}]$) are

$$\begin{aligned} L_{se}(t) = \rho UKB^2 \sum_{j=1}^n [(H_1^*(K)h_j(x) + H_2^*(K)\alpha_j(x) + H_5^*(K)p_j(x))\dot{\xi}_j(t)] \\ + \rho K^2 U^2 B \sum_{j=1}^n [(H_3^*(K)\alpha_j(x) + H_4^*(K)h_j(x) + H_6^*(K)p_j(x))\xi_j(t)]. \end{aligned} \quad (11)$$

Similar with equation (8), generalized modal self-induced force goes,

$$Q_{sei}(t) = \rho UKB^3 \sum_{j=1}^n c_{ij}^{se} \dot{\xi}_j(t) + \rho U^2 K^2 B^2 \sum_{j=1}^n k_{ij}^{se} \xi_j(t), \quad (12)$$

where c_{ij}^{se} and k_{ij}^{se} are the linear combination of flutter derivatives and the integral of modal shapes, the integral of modal shapes.

Regarding the total aerodynamic force as the superposition of quasi-steady buffeting force and unsteady self-induced force, the MDoF dynamic equation using normal coordinates reads,

$$\begin{aligned} [M]_{n \times n} \begin{Bmatrix} \ddot{\xi}_1(t) \\ \vdots \\ \ddot{\xi}_i(t) \\ \vdots \\ \ddot{\xi}_n(t) \end{Bmatrix}_{n \times 1} + ([C]_{n \times n} - [C^{se}]_{n \times n}) \begin{Bmatrix} \dot{\xi}_1(t) \\ \vdots \\ \dot{\xi}_i(t) \\ \vdots \\ \dot{\xi}_n(t) \end{Bmatrix}_{n \times 1} \\ + ([K]_{n \times n} - [K^{se}]_{n \times n}) \begin{Bmatrix} \xi_1(t) \\ \vdots \\ \xi_i(t) \\ \vdots \\ \xi_n(t) \end{Bmatrix}_{n \times 1} = \begin{Bmatrix} Q_{b1}(t) \\ \vdots \\ Q_{bi}(t) \\ \vdots \\ Q_{bn}(t) \end{Bmatrix}_{n \times 1}. \end{aligned} \quad (13)$$

where $[M]$ is the diagonal matrix of generalized modal mass; $[C]$ is the diagonal matrix of modal damping and $[K]$ is diagonal for structural stiffness. $[C^{se}]$ and $[K^{se}]$ are aerodynamic damping and stiffness matrices. The

asymmetric, and orthogonality no longer applies. The MDoF system can't be decoupled any more. It refers to the phenomenon of "aerodynamic coupling." The frequency response function (**FRF**) is then written in a matrix form as,

$$[H(\omega)] = [-[M]\omega^2 + i([C] - [C^{se}])\omega + ([K] - [K^{se}])]^{-1}. \quad (14)$$

The total modal response is calculated by the equation below.

$$\begin{aligned} \begin{bmatrix} S_{\xi_1 \xi_1} & \dots & S_{\xi_1 \xi_i} & \dots & S_{\xi_1 \xi_n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ S_{\xi_i \xi_1} & \dots & S_{\xi_i \xi_i} & \dots & S_{\xi_i \xi_n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ S_{\xi_n \xi_1} & \dots & S_{\xi_n \xi_i} & \dots & S_{\xi_n \xi_n} \end{bmatrix}_{n \times n} = [H(\omega)]^* \\ \begin{bmatrix} S_{Q_{b1} Q_{b1}} & \dots & S_{Q_{b1} Q_{bi}} & \dots & S_{Q_{b1} Q_{bn}} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ S_{Q_{bi} Q_{b1}} & \dots & S_{Q_{bi} Q_{bi}} & \dots & S_{Q_{bi} Q_{bn}} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ S_{Q_{bn} Q_{b1}} & \dots & S_{Q_{bn} Q_{bi}} & \dots & S_{Q_{bn} Q_{bn}} \end{bmatrix}_{n \times n} [H(\omega)]^T. \end{aligned} \quad (15)$$

After solving the modal response $S_{\xi_i \xi_j}$, SRSS and CQC methods can transfer the normal coordinate responses to physical responses. When the nondiagonal terms in the **FRF** matrix are neglected, it refers to the traditional mode-by-mode case, or SRSS.

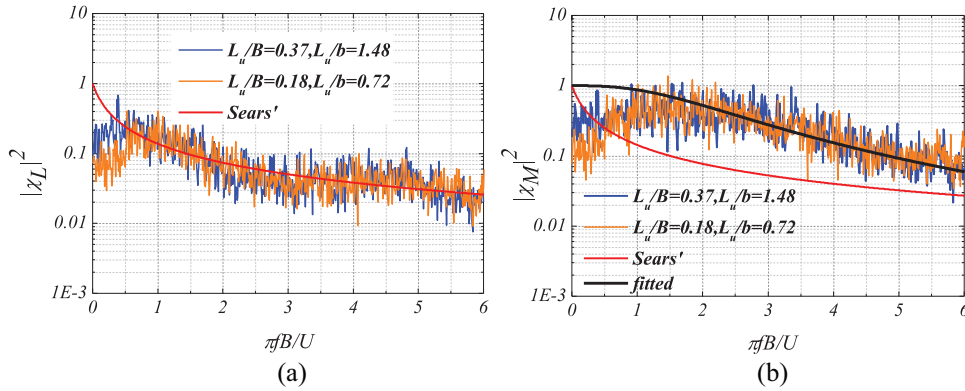


Figure 11. Tested AAFs of the bridge deck ($I = 14\%$): (a) lift force and (b) moment.

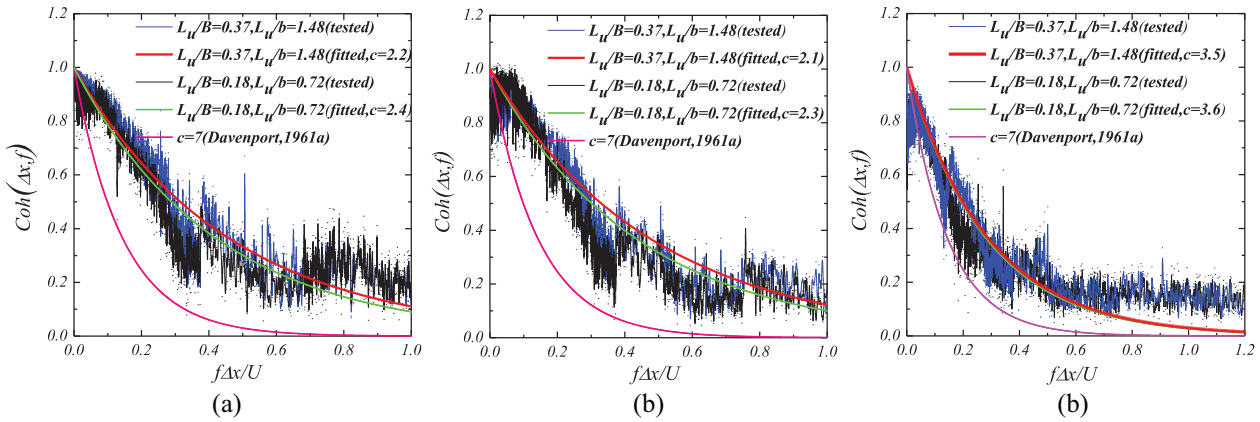


Figure 12. Tested span-wise coherence ($I = 14\%$): (a) lift force (c_1), (b) moment (c_2) and (c) cross term (c_3), L_u is integral scale.

Aerodynamic parameters of bridge deck

The aerodynamic parameters of present widely slotted section keep unrevealed. Another objective of present study is to check the applicability of analytical methods in extremely long cases. The accurate determination of aerodynamic parameters minimizes uncertainty, making the assessment reliable. Four sets of parameters are required, that is, static force coefficients, aerodynamic admittance functions, force coherences and flutter derivatives. Figure 4 shows the mean torsional responses keep smaller than 0.05° . Therefore, those parameters are determined when the attack angle is zero.

Mean coefficients. A sectional model of the same cross-section (Figure 10) was manufactured with the geometric scale ratio of 1:104. Totally seven span-wise sections were set on the model, and each section has 78 pressure taps. The details of model setup can be found in Xia et al. (2019). Pressure integration technique was conducted for the aerodynamic forces after the

correction of signal distortions (Bergh and Tijdeman, 1965). The mean forces were then normalized by the deck width B . As is shown, the drag coefficient is nearly an even function and the lift and moment coefficients are closer to zero at a zero angle, which is a reasonable result on a symmetric section. The data in Figure 10 was obtained under smooth flow, whereas, the grid turbulence in the picture is for the data in Figures 11 and 12.

Admittance functions. A common understanding of aerodynamic admittance function (e.g. Larose et al., 1998; Li et al., 2018) is when the scale is sufficient, most sections' AAF will approach the Sears function, which is theoretically deduced on an airfoil. However, this common sense is in the background of single-box girders. It is unrevealed whether this tendency still applies on a twin-box girder. Grid turbulence (Figure 10) was used with the turbulence intensity the same as that in the full aeroelastic model testing, that is, $I = 14\%$.

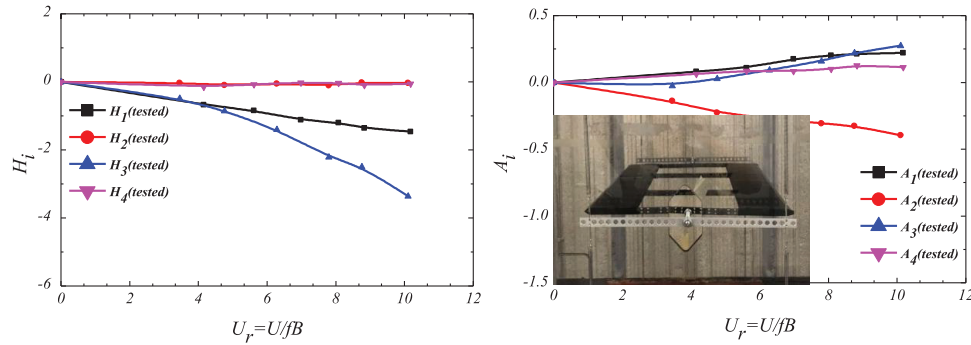


Figure 13. Aerodynamic derivatives of the bridge deck.

The drawback of grid turbulence is the insufficiency of integral scales since they are of the order of grid sizes. Its infeasible to make the integral scales proportional to the aeroelastic model testing. This insufficiency will cause deficit of the AAF, and the deficit is confined in the low frequency region (Larose et al., 1998). To assess this effect, two integral scales were tried in Figure 11. It shows for lift AAF, the deficit is restricted in the region of $\pi f B/U < 0.5$ and for moment is $\pi f B/U < 1.0$. With the increasing of integral scales (from orange line to the blue), the lift AAF is enlarged in the low frequency range and becomes close to the Sears function. An extreme situation of $f = 0$ means the turbulence size is super large and fluctuating surface pressure changes synchronously so that no reduction is expected. In other words, the AAF at $f = 0$ must be 1.0. Thus, we made a logical extrapolation that the deviation from the Sears function in Figure 11(a) is caused by the insufficient integral scale in the section model testing and lift AAF can be replaced by Sears function (Sears, 1941). From Figure 11(b), a similar tendency can be found in the low frequency range, namely the AAF increases with the integral scales and should be 1.0 when $f = 0$. However, its variation in the high frequency range is highly disparate from the Sears function. Therefore, the AAF of moment is mathematically fitted. The reason why the moment AAF is utterly different from the Sears function will be explained in Figure 17.

Span-wise coherence of fluctuating force. While the AAFs corrects the in-plane scales' effect, an out-plane reduction is also expected, which is known as span-wise coherence. Figure 12 is from the seven sections on the sectional model in Figure 10 which shows the relation of the root coherence and the reduced frequency $f \Delta x/U$. Like the coherence of wind fluctuations, buffeting forces' coherences also can be fitted by exponential laws. In the early work, the force coherence is replaced by those of upstream wind. Davenport (1961) used a

decay factor $c = 7$. However, higher force coherences have been pointed out on single box girders (i.e. Sankaran and Jancauskas, 1993; Larose et al., 1998). The sectional testing found the decay factor of wind was around 7.0 to 10.0 (Xia et al., 2019) while the aerodynamic forces decay much slower. The present deck's decay factors of self-terms are in the range of 2.0 to 2.5 and around 3.5 for the cross term.

Similarly, turbulence scale is a controlling factor for span-wise coherence. An increased ratio of turbulence scale to deck width enhances the coherence (Larose et al., 1998). Since the insufficiency of turbulence scale is inevitable using grid-turbulence, a comparison of two grid configurations in Figure 12 shows the variation of decay factors. The effect of turbulence scale is in some sense, small. In the full aeroelastic model testing, the L_u/B is 0.53 ($L_u/b = 2.12$). Its infeasible to make the integral scales proportional to the aeroelastic model testing. An empirical method was proposed in Xia et al. (2019) which realizes the modification of span-wise force under different turbulence conditions.

Aerodynamic derivatives. Another sectional wind tunnel testing focusing on 2D dynamic response was conducted in Figure 13. A modified nonlinear least square method realizes derivatives' identification using the free vibration data (Li et al., 2003). Figure 13 is the variations of flutter derivatives with reduced wind speed. During the least square fitting, six cycles of response data after short-time impulse were used. Those time histories calculated by flutter derivatives can match well with the tested ones.

Assessment of methods' applicability

To assess the applicability of classical methods in a 5000 m case, four methods are employed, as is given by Figure 14(a). The measured modal shapes, tested wind spectra and aerodynamic parameters are used for calculation. Considering the limited contribution of

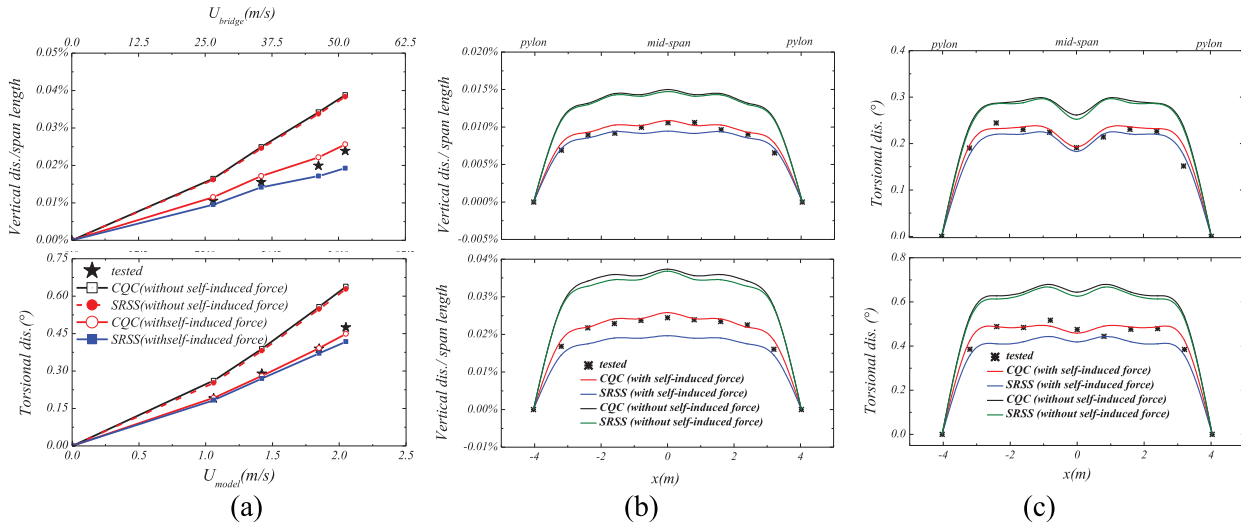


Figure 14. Assessment of buffeting methods: (a) RMS. response at mid-span, (b) Span-wise RMS. responses, $U_{\text{model}} = 1.1$ m/s, $U_{\text{bridge}} = 26.4$ m/s; (c) Span-wise RMS. responses, $U_{\text{model}} = 2.1$ m/s, $U_{\text{bridge}} = 51.1$ m/s.

higher modes, the calculation has taken the four vertical and four torsional modes into account, which means first two symmetric and two anti-symmetric modes. The results obtained are compared to the measurements from full aeroelastic model testing. Since only symmetric modes have their participations at mid-span in Figure 14(a), a more comprehensive comparison is given in Figure 14(b) and (c) at two typical speeds.

The former two methods take no self-induced force into account so that only structural coupling effect is considered for CQC combination. The difference of these two methods shows structural coupling effect is small. It is because the modal frequencies are well separated on a 5000 m bridge. When the latter two methods are conducted, the responses become much smaller. Unsteady effect is a stable factor which causes a reduction of 40% to 50%. The increasing of wind speed amplifies this effect. In general, vertical response is more sensitive to self-induced force. This can be explained with a glimpse of the flutter derivatives in Figure 13 and the definition in equation (10). The main contributor of self-induced force is caused by the movement along its own direction, namely the H_1^* and H_4^* terms for vertical motion, and A_2^* and A_3^* terms for torsional displacement. H_1^* is related to vertical aerodynamic damping and H_4^* to aerodynamic stiffness. Both H_1^* and H_4^* have a negative number, which means positive aerodynamic damping and stiffness suppressing the responses. However, A_2^* and A_3^* have opposite signs with each other. The unsteady effect along torsional direction is then partially offset.

The results from mode-by-mode (SRSS) method refers to the case where no cross term in the **FRF**

matrix is considered. It gives a smaller result than the tested ones. As Chen et al. (2000) mentioned, the aerodynamic coupling effect will be more prominent at a higher wind speed. This phenomenon can be understood if one looks at the PSD of mid-span response in Figure 15. The traditional mode-by-mode method only simulates the unsteady effect but neglects the aerodynamic coupling, thus, is risky for buffeting analysis.

In summary, Figure 14 shows the CQC methods with self-induced forces can predict the total response amplitudes while Figure 15 illustrates this traditional method can precisely reconstruct the spectrum and kinetic energy in a 5000 m case. The small deformation and linear assumption keep applicable for buffeting analysis though the structural nonlinearity could be highlighted during static design of super-long bridges.

Parameters sensitivity discussion

The implementation of buffeting calculation requires plenty of aerodynamic parameters. The aerodynamic parameters are partially assumed in most other researches (e.g. Chen et al., 2000), although in present study these parameters are experimentally decided. A sensitivity discussion helps understand which parameters are the most important and their effect.

Span-wise coherence. The results in Figures 14 and 15 are calculated using the tested force coherences. On the other hand, the recommended value by Davenport, $c = 7$, is also used in Figure 16(a). Since aerodynamic forces are more coherent than inflow, buffeting responses are highly underestimated if the

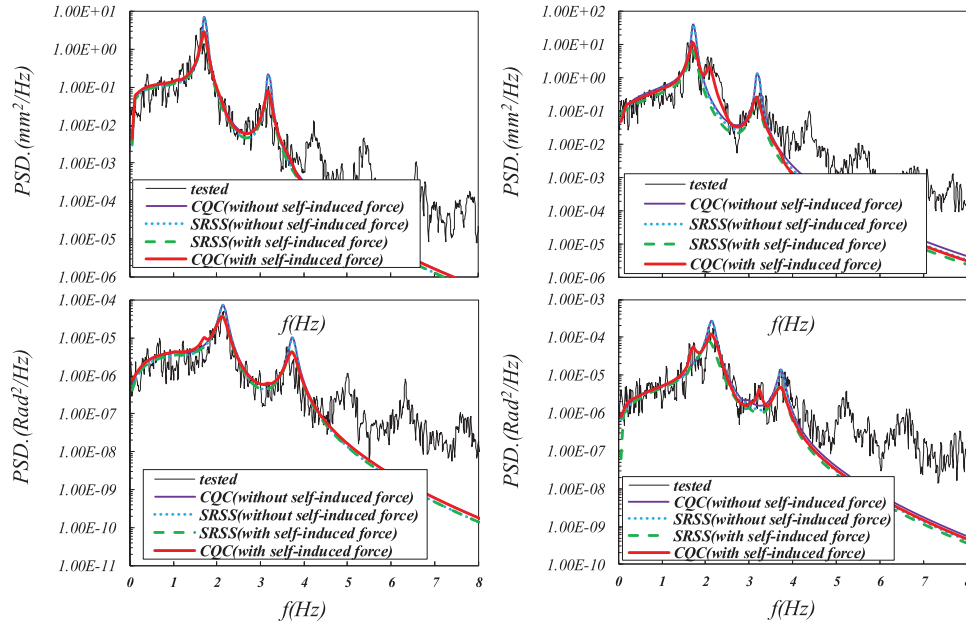


Figure 15. Assessment of methods in the frequency domain: (a) $U_{\text{model}} = 1.1$ m/s, $U_{\text{bridge}} = 26.4$ m/s and (b) $U_{\text{model}} = 2.1$ m/s, $U_{\text{bridge}} = 51.1$ m/s.

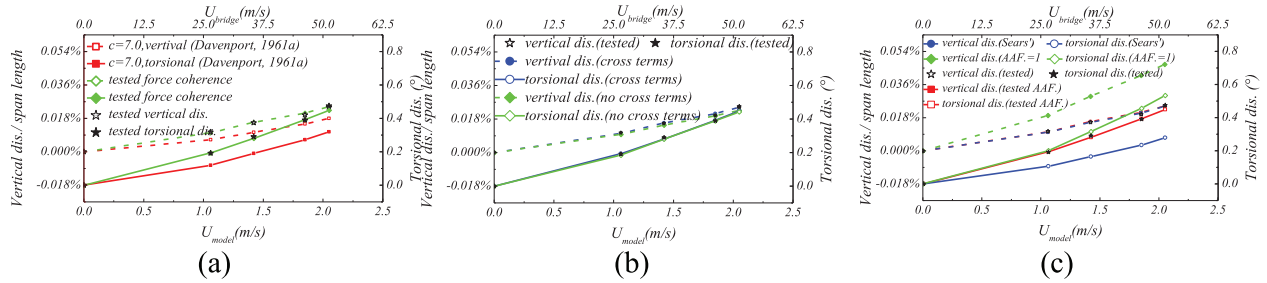


Figure 16. RMS. responses using different: (a) coherences, (b) cross terms, and (c) AAFs.

recommended factors are used. This deviation is amplified at a higher wind speed. For instance, the torsional response is only 60% of that by the tested coherence when experimental wind speed is 2.1 m/s. The results using the tested decay factors are closer to the measured responses. From another perspective, the span-wise coherence function previously tested is reasonable.

Cross terms of modal loadings. A large proportion of computing time is consumed for the double integrals in JAF. It will be time-efficient if the cross term $S_{b_{ij}}$ can be ignored ($i \neq j$). The cross spectra of modal buffeting force is set to be zero manually so that the second matrix in the right-hand side of equation (15) becomes diagonal. Figure 16(b) shows the neglect slightly underestimates the buffeting response. Compared with

the influence of self-induced force, aerodynamic admittance functions in the later section and force coherence, this impact is negligible.

Aerodynamic admittance function. The above results are obtained by the tested AAF in Figure 11. Buffeting responses when AAFs are both set to be 1.0 or Sears' function are compared in Figure 16(c). As what Scanlan did in his early work (1978), the neglect of upstream turbulence's scale and AAF provides an overlarge safety factor. The safety factor of a 5000 m bridge along vertical direction is the order of two to three. The tested lift AAF is close to Sears' function so that the calculated vertical responses match the measured ones. On the other hand, the moment AAF is larger than Sears' function. If Sears' function is used, it will cause a considerable underestimation especially

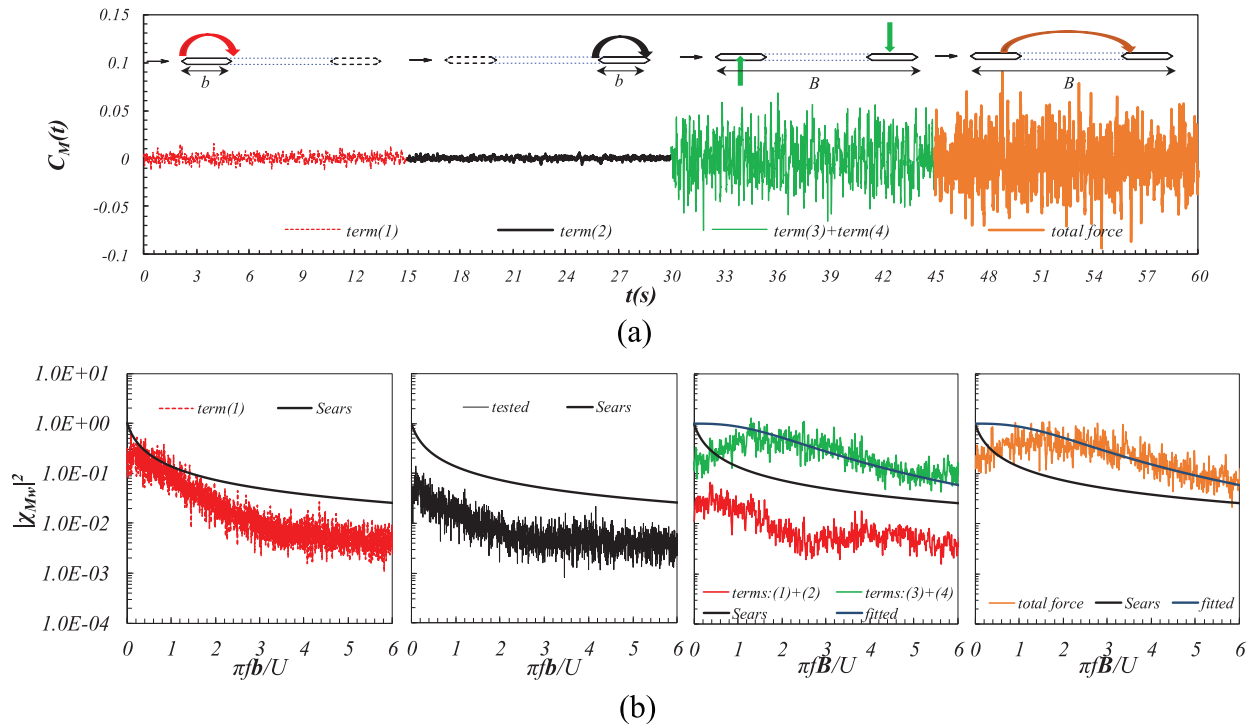


Figure 17. The contribution of separated decks: (a) typical time histories and (b) separated AAFs.

at high wind speeds. The torsional underestimation is the order of 50%.

Larger torsional responses are presented in Figure 5 compared with the other bridges. A preliminary study shows their force coefficients are comparable. Based on the discussion in Figure 16, the most important factor is aerodynamic admittance function. When calculating the torsional displacement, the tested force coherence and AAF in Figure 11 are used. In Figure 16(c), the torsional response will be significantly reduced once the Sears' function is used, making the data comparable with the existing bridges. It is generally accepted that for a single box girder like the Nansha Bridge in Figure 5, the moment AAF will approach Sears' function. Therefore, we think the excessive torsional response of the present deck in Figure 5 is caused by the oversized torsional AAF compared with the Sears' function.

Figure 17(b) gives the AAFs from the sectional model pressure testing. The data on the windward and the leeward deck leads to the AAFs on the corresponding side box. When the separated AAFs are calculated, the horizontal axis is given by the width of one box, b , while for the whole deck, the total width B is used. The AAF on the windward deck is close to the Sears' function, which indicates the previous pressure testing for AAF is reliable. However, for the leeward deck, the separated AAF is an order of magnitude smaller.

This is understandable because the windward deck blocks the upstream wind and reduces the overall wind loading. When the fluctuating forces on the both decks are put together and B is used, it yields the total AAF larger than Sears function in Figure 11.

The above discussion illustrates the overlarge AAF distribution along torsional direction as the result of deck configuration. This kind of distribution can also be found on the Xihoumen Bridge which is also central-slotted (Wen, 2008). For this kind of central-slotted deck, the total moment can be divided into four parts, as (1) the windward deck's torque to its own center, (2) the leeward deck's own torque, (3) the moment caused by the windward lift to the whole deck's center, and (4) the leeward lift's contribution.

A typical time history of aerodynamic via sectional pressure testing is given in Figure 17(a) which leads to the AAFs in Figure 17(b). For clarity, 15 s data are plotted although the AAFs are calculated using 120 s data. The latter two terms, especially the moment caused by the windward lift, are the main contributors of the total torque. This term is the product of lift force and deck width so that the large deck width amplifies this effect for moment AAF instead of lift AAF. The occurrence of central slot adds coherent turbulent structure onto the leeward deck. The corresponding admittance function is larger than Sears' function. In some sense, this effect also makes the torsional

displacement on Xihoumen Bridge larger than Nansha Bridge of a similar span-length in Figure 5.

Conclusion

To understand the buffeting performance of super-long suspension bridges, the full aeroelastic model testing of a 5000 m span-bridge has been presented accompanied with the identification of aerodynamic parameters. Meanwhile, frequency-domain calculations have been carried out to check applicability of classical methods in super-long cases. The following conclusions can be made:

- (1) For a 5000 m bridge, buffeting is an important contributor to the total vertical and torsional responses while the lateral response is quasi-static. The lateral displacements are irrelevant with torsional and vertical response based on responses' PSD. When the wind speed is 51.1 m/s, the mean lateral response can be controlled to the 1/500 of span length. The vertical RMS buffeting response on a 5000 m bridge is the order of 1/10000 to 1/5000 of span length and the torsional RMS values approaches 0.45° . Lateral and vertical responses are comparable with Xihoumen Bridge and Nansha Bridge while the torsional response is larger. A 5000 m bridge's peak factors are from 3.0 to 4.0, lower than the recommended range.
- (2) More than 90% of kinetic energy is provided by the 1st mode. First two orders of modes are enough for buffeting analysis since higher orders' participation is $<3\%$. Vertical buffeting kinetic energy is more concentrated than the torsional one. Structural coupling effect is negligible but with an increased wind speed, aerodynamic coupling effect is prominent. Aerodynamic coupling effect accounts for 20% of total energy when the wind speed approaches 50 m/s. Besides, signature turbulence's contribution is negligible in present case.
- (3) Self-induced force is an important safety factor. It makes the buffeting response 40% to 50% lower than no self-induced force occasions. Meanwhile, it affects the vertical response more than the torsional. Both SRSS and CQC results without self-induced force are conservative for a 5000 m bridge. After taking self-induced force into account, the multi-mode buffeting analysis method can still be used.
- (4) Replacing the force decay factor by wind makes the computed torsional response is only

60% of the real value at an increased wind speed. It is acceptable to omit the cross terms of modal buffeting force. Aerodynamic admittance function is more important than other parameters. Sears' function is risky for torsional response while the employment of a unit AAF is conservative. The larger torsional displacement than other bridges is caused by the overlarge moment AAF resulting from the wide central slot.

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